The 3-adic mirror of the reduced Collatz dynamics: duality, dead ends, and a density bound

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Abstract

The companion paper [1] compresses the Collatz dynamics into a self-map F on states (ω, d) and shows the forward per-step arithmetic is governed by a 2-adic logarithm, the anchor $M(\omega)$. Here we run the map backward. The predecessor structure of F is characterized completely: predecessors arrive through the *doors* (representatives) of a state, one per admissible exponent s, and the backward depth obeys the exact law $d = 1 + v_3(s - M_3(y))$, where the backward anchor $M_3(y)$ is an affine 3-adic logarithm base 2 whose additive unit $M_3(1)$ is the 3-adic realization of -1. The entire forward per-step theory then dualizes under the exchange $2 \leftrightarrow 3$, $s \leftrightarrow d$, with four precisely identified asymmetries rather than forced analogies: the backward digit flow has one fewer cross-prime step; the backward increment law is partial, failing exactly on the known door-mortality set, where the forward law is total; the forward decision trichotomy degenerates backward to a dichotomy, because $(\mathbb{Z}/3^q)^\times$ is cyclic while $(\mathbb{Z}/2^q)^\times$ is not; and the vertical ladder laws on both sides are gated by the respective anchors, with a forced step-size difference. We map the dead ends of backward generation: exactly one door per state is mortal, dead for exactly half of all states, and the states with no predecessors at all are precisely those of the odd multiples of 3 at depth one. Steering laws are proved: backward branching steers the 3-adic coordinate completely, freezes the 2-adic residues at 3^{-d}, and consequently places the predecessor's forward anchor on the exact lattice $-d\,M(3)$ to any prescribed precision. Finally, a single-type sub-tree (via a collapse identity making the door map independent of predecessor depth) yields a fully self-contained lower bound: at least $2^{-3.3}X^{0.3}$ odd integers below X reach 1, with critical exponent 0.3304 for this core. The bound is far weaker than the linear-programming results of Krasikov–Lagarias ($X^{0.84}$); its contribution is the derivation route, in which the classical difference *inequalities* are replaced by an exact local law, and a mapped refinement path.

1 Introduction

Fix the odd-to-odd Collatz map $T(x) = (3x+1)/2^{v_2(3x+1)}$. The companion paper [1] introduces reduced states (ω, d) (ω odd, $3 \nmid \omega$, $d \geq 1$), a self-map F whose convergence problem is exactly equivalent to the Collatz conjecture, and proves that the forward per-step arithmetic is governed by the 2-adic anchor $M(\omega) = -2\log\omega/\log 9 \in \mathbb{Z}_2$: the exit valuation obeys $s = 2 + v_2(d - M(\omega))$ on the lifting classes, the depth evolution closes exactly, and a finite window of digits decides each forward step without error.

*`macindoebenjamin@gmail.com`. Companion to [1], whose coordinate system and forward laws are assumed. Developed through AI-assisted research (Claude, Anthropic; one delegated session by Claude Sonnet, independently reviewed); responsibility and verification protocol as stated in [1], Appendix A, and in Appendix A below.

This paper asks what the same machinery looks like in reverse, where the deterministic map becomes an infinitely-branching tree. The short answer, developed in Sections 3–5, is that everything dualizes, with the primes 2 and 3 exchanging roles: backward arithmetic is 3-adic, governed by an anchor M_3 which is an affine discrete logarithm base 2, through a valuation law that mirrors the forward one term for term. The longer answer is that the mirror is not a formality — each dual theorem carries a structural asymmetry with an identifiable cause, and the asymmetries are where the content lives.

Three applications follow. Section 6 maps the *dead ends* of backward generation — where the tree stops growing — and finds mortality remarkably concentrated: one door per state, dead half the time by an explicit parity law, with total barrenness only at the states of odd multiples of 3 at depth one. Section 7 proves *steering* laws: what the infinite backward branching can and cannot control. It controls the 3-adic data completely; it cannot move the 2-adic residues at all (they freeze at 3^{-d}); and through the freeze it places the predecessor's *forward* anchor on an exact lattice — so anchor variation, which forward is the central unsolved difficulty of [1], is backward a controlled linear walk. Section 8 extracts a fully rigorous density bound from a deliberately impoverished sub-tree: at least $2^{-3.3}X^{0.3}$ odd integers below X reach 1 (Theorem 8.4). The bound is honest about its position — well below Krasikov's 1989 exponent 0.43 and far below the Krasikov–Lagarias 0.84 [3] — and about what it offers instead: a derivation in which the local branching is known *exactly* rather than bounded by difference inequalities, and a refinement path whose stages correspond to the resources the base theorem discards.

Contributions. (i) Complete characterization of F^{-1} (Theorem 3.1). (ii) The backward anchor M_3, its algebra, and the backward valuation law (Theorem 4.3), with the backward branching ledger $P(d = j) = 2 \cdot 3^{-j}$. (iii) Dualization of the forward per-step theory with four identified asymmetries (Section 5). (iv) The dead-end classification: door mortality and Gardens of Eden (Theorems 6.1, 6.2). (v) Steering laws, including the backward anchor-placement law $M(\omega_{\text{pred}}) \equiv -d\, M(3) \pmod{2^{s-2}}$ (Theorem 7.1). (vi) A rigorous density bound with explicit constants (Theorem 8.4), via a collapse identity that reduces the counting core to a single type.

Related work. Predecessor trees for the $3x+1$ problem are classical: Wirsching's monograph [4] develops them systematically, and quantitative lower bounds on $\pi_1(x) = \#\{n \le x : 1 \in \text{orbit}(n)\}$ run from Crandall [2] through Krasikov's difference inequalities to the computer-assisted $x^{0.84}$ of Krasikov–Lagarias [3]; see Lagarias [5] for the literature. The verified range of the conjecture is 2^{71} [6]. A contemporary Lean-formalized conditional treatment of cycles is due to Merle [7]. Relative to this literature, the present paper's predecessor structure is stated on reduced states rather than integers, its local law (Theorem 4.3) is exact rather than an inequality, and its density bound is chosen for self-containedness rather than strength.

2 Preliminaries from the companion paper

For an odd x write $x + 1 = 2^m 3^a \omega$ with $3 \nmid \omega$; the projection is $R(x) = (\omega, m + a)$. For a state (ω, d): $A = 3^d\omega - 1$, $s = v_2(A)$, $x_{\text{exit}} = A/2^s$, $C = A + 2^s$, $\sigma = v_2(C)$, $a_+ = v_3(C)$, and $F(\omega, d) = \big(C/(2^\sigma 3^{a_+}), (\sigma - s) + a_+\big)$. The representatives of (ω, d) are $y_a = 2^{d-a}3^a\omega - 1$ for $0 \le a \le d - 1$; the block from any representative exits at x_{exit}, and the Collatz conjecture is equivalent to every F-orbit reaching $(1, 1)$ [1, Thm. 2.3]. The forward anchor is $M(\omega) = N(\omega^2)$ where $9^{N(u)} = u^{-1}$ in $\mathbb{Z}_2$; on lifting classes ($3^d\omega \equiv 1 \bmod 8$), $s = 2 + v_2(d - M(\omega))$ [1, Thm. 3.4].

3 The predecessor structure

Theorem 3.1 (complete characterization of F^{-1})**.** *Let (Ω, D) be a valid state with representatives (*doors*) y_a. The predecessors of (Ω, D) are exactly: for each door $y = y_a$ with $3 \nmid y$, and each $s \geq 1$ with s odd if $y \equiv 1 \pmod 3$, even if $y \equiv 2 \pmod 3$, the state*

$$(\omega, d) = \big(N/3^{v_3(N)},\ v_3(N)\big), \qquad N = 2^s y + 1,$$

where $3 \mid N$ automatically. Distinct (a, s) give distinct predecessors, and there are no others. Doors with $3 \mid y$ (possible only at $a = 0$) admit no predecessors.

Proof. $F(\omega, d) = (\Omega, D)$ iff $x_{\text{exit}}(\omega, d) = y$ for some door y, i.e. $3^d\omega = 2^s y + 1$ with $s = v_2\big(3^d\omega - 1\big)$. Given (y, s), factoring $N = 3^d\omega$ with $3 \nmid \omega$ forces $d = v_3(N)$ and $\omega = N/3^d$; the divisibility $3 \mid N$ holds iff $2^s y \equiv 2 \pmod 3$, the stated parity condition; and $v_2\big(3^d\omega - 1\big) = v_2(2^s y) = s$ exactly since y is odd. (Verified against brute-force forward scans on seven targets: exact agreement.) $\square$

4 The backward anchor and valuation law

The unit group mod 3^k is cyclic of order $2 \cdot 3^{k-1}$, generated by 2; exponents live in $\mathbb{Z}/2 \times \mathbb{Z}_3$.

Definition 4.1. *For odd y with $3 \nmid y$, let $M_3(y) \in \mathbb{Z}/2 \times \mathbb{Z}_3$ solve $2^{M_3(y)} = -1/y$. Its parity component is determined by y mod 3 (and matches the parity condition of Theorem 3.1); its $\mathbb{Z}_3$ component is the backward anchor proper.*

Proposition 4.2 (algebra of M_3)**.** *$M_3(1)$ is the 3-adic realization of -1: $2^{3^{k-1}} \equiv -1 \pmod{3^k}$, so $M_3(1) \equiv 3^{k-1} \pmod{2 \cdot 3^{k-1}}$. Moreover $M_3(y_1y_2) = M_3(y_1) + M_3(y_2) - M_3(1)$: M_3 is the affine logarithm $M_3(y) = M_3(1) - \log_2 y$ in the 3-adic discrete logarithm. (Verified at depth 3^8; 300 random pairs, zero failures.)*

Theorem 4.3 (backward valuation law)**.** *For every live door y and every s of the correct parity,*

$$d \ = \ v_3(2^s y + 1) \ = \ 1 + v_3(\, s - M_3(y)\,)\,.$$

Proof. $2^s y + 1 = y\, 2^{M_3(y)}\big(2^{s - M_3(y)} - 1\big)$ since $2^{-M_3(y)}y^{-1} = -1$; the prefactor is a 3-adic unit; the parity condition makes $s - M_3(y)$ even, and the lifting-the-exponent isometry $v_3\big(2^t - 1\big) = 1 + v_3(t)$ (even t) applies. (Verified: 4,265 random checks at anchor depth 3^8, zero failures.) $\square$

Corollary 4.4 (backward ledger)**.** *Along any door's branch family, the exact density of s with $d = j$ is $2{\cdot}3^{-j}$: the mirror of the forward ledger $P(s = j) = 2^{-j}$. (Measured $0.6664, 0.2230, 0.0736, 0.0245, 0.0082$ against $2/3, 2/9, 2/27, 2/81, 2/243$.)*

Table 1 summarizes the duality; note the exchange is exact in structure and inexact in constants, and the inexactness is itself explained (Section 5).

5 The dual per-step theory, with its asymmetries

Each per-step theorem of [1, §3] has a proved 3-adic mirror. We state them compactly with their verification counts, emphasizing in each case the asymmetry — the place where the mirror is *not* a translation, and why.

forward [1]	backward (this paper)
arithmetic prime 2	arithmetic prime 3
$s = v_2\big(3^d\omega - 1\big)$	$d = v_3(2^s y + 1)$
$M(\omega) = -\log\omega^2/\log 9 \in \mathbb{Z}_2$	$M_3(y) = M_3(1) - \log_2 y \in \mathbb{Z}/2 \times \mathbb{Z}_3$
$s = 2 + v_2(d - M(\omega))$ (lifting classes)	$d = 1 + v_3(s - M_3(y))$ (all live doors)
$v_2(9^t - 1) = 3 + v_2(t)$	$v_3(2^t - 1) = 1 + v_3(t)$
ledger 2^{-j}	ledger $2 \cdot 3^{-j}$
classes mod 8 gate the law	class mod 3 gates the parity
deterministic orbit	infinitely-branching tree
equidistribution hypothesis	tree density in $\mathbb{N}$

Table 1: The $2 \leftrightarrow 3$ duality.

Theorem 5.1 (mirror digit-determinacy). *$M_3(y) \bmod 3^k$ is determined by $y \bmod 3^{k+1}$; $N = 2^s y{+}1 \bmod 3^q$ by $y \bmod 3^q$ and $s \bmod 3^{q-1}$ (parity fixed); and $\omega = N/3^d \bmod 3^r$ by $N \bmod 3^{d+r}$. Chaining these, a 3-adic window on (y, s) of depth $W \geq d + r$ determines the predecessor's $\omega \bmod 3^r$ exactly.* Asymmetry: *the forward chain needs one genuinely cross-prime step (the order of 3 mod 2^r, to strip a 3-power from a 2-adically analyzed quantity); the backward chain needs none, because N is odd by construction and the only removal is same-prime. (3,000 checks per fact; $\approx$ 2,670 per window depth; zero failures.)*

Theorem 5.2 (backward increment law, and its freeze). *Write $y_0(\kappa, K) = 2^K\kappa - 1$ for a state's $a{=}0$ door. Across a backward step through door y at branch s, the increment $M_3(y') - M_3(y)$ of top-door anchors is determined mod 3^k by $y \bmod 3^{d+k+1}$ and $s \bmod 3^{d+k}$* — whenever y' is alive. Asymmetry: *the forward increment law of [1] is total; the backward one is partial, undefined exactly on the door-mortality set of Theorem 6.1, i.e. on precisely half of top-door lineages. The failure is hard: no window depth resolves it, because the branch is absent, not hidden. (6,000 trials; freeze rate 0.5059 against the exact $\frac{1}{2}$; window recovery on all alive cases, zero failures.)*

Theorem 5.3 (one-step dichotomy). *From a depth-K window on (y, s) alone: either $(s - M_3(y)) \bmod 3^K \neq 0$ and the predecessor depth is exact, or the window correctly reports $d \geq K{+}1$; undecided rate $\approx 3^{-K}$.* Asymmetry: *the forward analogue is a trichotomy — six of eight residue classes get $s \in \{1, 2\}$ as class constants at zero window cost. Backward there is no free branch: Theorem 4.3 is unconditional across all live doors. The cause is group-theoretic: $(\mathbb{Z}/3^q)^\times$ is cyclic, while $(\mathbb{Z}/2^q)^\times \cong \mathbb{Z}/2 \times \mathbb{Z}/2^{q-2}$ has the extra factor that furnishes the forward theory's zero-cost classes. ($\approx$ 13,200 trials per depth $K \in \{2, 4, 6, 8\}$; zero decision errors; zero deep-bound violations.)*

Theorem 5.4 (the two ladders). ***Forward (vertical):*** *for fixed ω, $A(\omega, d{+}1) = 3A(\omega, d) + 2$ forces: if $s(\omega, d) = 1$ then $x_{\text{exit}}(\omega, d{+}1) = T(x_{\text{exit}}(\omega, d))$ — the two states' futures are the same integer orbit, offset one step; if $s \geq 2$, the exits split by the kick $e \mapsto 3 \cdot 2^{s-1}e + 1$, and $s(\omega, d{+}1) = 1$. Since $s = 1$ occupies alternating depths, every column of the state space is a strict braid of Collatz steps and kicks, tearing exactly at the anchor's spikes. (30,000 states, zero failures both branches.)* ***Backward (dual):*** *for a fixed door y, $N(y, s{+}2) = 4N(y, s) - 3$ forces: if $d(y, s) = 1$ then $\omega(y, s{+}2) = T_3(\omega(y, s))$ with $T_3(\omega) = (4\omega - 1)/3^{v_3(4\omega - 1)}$; if $d \geq 2$ then $\omega(y, s{+}2) = 4 \cdot 3^{d-1}\omega - 1$ exactly, with $d(y, s{+}2) = 1$. The tear-line is gated by the first digit of the backward anchor, mirroring the forward gate.* Asymmetry: *the coefficient is $4 = 2^2$, not 3: the parity condition confines s to a lattice of index 2, so the dual ladder's unit step is forced to be 2. (19,992 trials, zero failures both branches.)*

6 Dead ends

Theorem 6.1 (door mortality)**.** *Doors y_a with $a \geq 1$ satisfy $y_a \equiv 2 \pmod 3$ identically: they are never dead. The sole mortal door is $a = 0$, dead iff $2^D\Omega \equiv 1 \pmod 3$ — exactly one combination of $(\Omega \bmod 3,\ D \bmod 2)$, i.e. half of all states. (20,000 states, all doors, zero failures.)*

Theorem 6.2 (Gardens of Eden)**.** *A state has no F-preimage iff $D = 1$ and $\Omega \equiv 2 \pmod 3$ — equivalently, iff its unique representative is an odd multiple of 3. Every state with $D \geq 2$ is reachable.*

Classically, one third of odd numbers are unreachable *values*; in reduced coordinates the phenomenon concentrates: deeper states merely lose one door, and total barrenness occurs only at depth one. A mortality-corrected branching (Malthusian) analysis of the full tree — using the measured stationary depth law as its one non-rigorous input — yields a renewal mass exceeding 1 for every exponent $c < 1$ (minimum ≈ 1.52 at $c \approx 0.7$): the tree is supercritical at every sub-density exponent, predicting full density, consistent with exact enumeration of the tree to $\omega \leq 2^{19}$ (408,302 states, empirical exponent 0.97–0.98 and rising). This paragraph is labeled heuristic; the rigorous extraction is Section 8.

7 Steering

What does infinite backward branching control? Fix a live door y and sweep s.

Theorem 7.1 (steering laws)**.** *(i)* Depth: total control. *For any target d^*, the branches with $d = d^*$ form an exact-density-$2 \cdot 3^{-d^*}$ subset of the allowed s (Theorem 4.3); every depth is hit infinitely often.* *(ii)* 2-adic residues: frozen. *As $s \to \infty$, $N = 2^s y + 1 \to 1$ in $\mathbb{Z}_2$, so the predecessor's residues freeze: $\omega \equiv 3^{-d} \pmod{2^{\min(s,k)}}$. Direct 2-adic steering is impossible; only finitely many small-s branches deviate. (Empirically: from one door, 6 of 64 windows $(\omega \bmod 32, d \leq 4)$ are reachable at all.)* *(iii)* Forward-anchor placement: complete control through the depth. *Consequently, for s in the stable range,*

$$M(\omega_{\text{pred}}) \equiv -d \cdot M(3) \pmod{2^{s-2}},$$

and since $M(3)$ is odd (a 2-adic unit), the lattice $\{-d\,M(3)\}$ covers every residue: choosing d (steerable by (i)) and s large places the predecessor's forward anchor anywhere modulo 2^k, to arbitrary precision. (2,025 random checks of (iii) at depth 20, zero failures, with the depth bound $s - 2$ attained exactly.)

Proof of (iii). $\omega\, 3^d = 1 + 2^s y$, so $\omega^2 9^d = (1+2^s y)^2 = 1 + 2^{s+1}y + 2^{2s}y^2$, whence $M(\omega) + d\,M(3) = N\big((1+2^s y)^2\big)$, whose 2-adic valuation is $s - 2$ by the forward isometry. Oddness of $M(3)$ is the parity statement of [1, §3] at $\omega = 3$. □

Remark 7.2. Forward, the anchor increment along an orbit is the central unresolved object of [1] (the "fiber-to-orbit bridge"). Backward, by (iii), the anchor walk is *controlled and linear.* The same walk changes epistemic status with direction of travel; whether the backward controllability can be turned into forward information is, in our judgment, the sharpest open question this paper raises.

8 A rigorous density bound

Definition 8.1 (door tree)**.** *Root $y = 1$. Children of a node y (odd, $3 \nmid y$): for each allowed s (parity per y mod 3), $y' = (2^{s+1}y - 1)/3$, kept when $3 \nmid y'$ and $y' > 1$.*

Lemma 8.2 (collapse identity)**.** *Every kept y' is the designated door of a genuine F-predecessor of y's state: the depth-1 predecessor's door $2\omega' - 1$ and the depth-d ($d \geq 2$) predecessor's door $2 \cdot 3^{d-1}\omega' - 1$ are* the same expression $(2^{s+1}y - 1)/3$ *— the door map is independent of the predecessor's depth. (13,408 checks, both types, zero failures.)*

Lemma 8.3 (triple law)**.** *For any three consecutive allowed s, the values $2^s y + 1$ are $\equiv \{0, 3, 6\} \pmod 9$, one each (they differ by steps of $3 \cdot 2^s y \equiv 6 \bmod 9$). Hence the three candidate y' are $\equiv \{0, 1, 2\} \pmod 3$, one each: exactly two children per triple are kept. (20,000 checks.)*

Theorem 8.4 (density bound)**.** *Let $\tilde{\pi}(X) = \#\{$odd $y \leq X$: the T-orbit of y reaches $1\}$. Then for all $X \geq 1$,*

$$\tilde{\pi}(X) \;\geq\; 2^{-3.3}\, X^{0.3},$$

and the argument yields any exponent c with $(2^{-3.415c} + 2^{-5.415c})/(1 - 2^{-6c}) > 1$; the critical value is $c^* \approx 0.3304$.

Proof. Nodes of the door tree are doors of states backward-reachable from $(1, 1)$, hence (equivalence theorem of [1]) integers whose orbits reach 1; they are pairwise distinct (a door determines its state, states in the backward tree are distinct, and the two child types are separated mod 3). Each kept child multiplies its parent by less than $2^{s+1-\log_2 3}$, so the branch at s has log-increment $\delta(s) < s - 0.585$. By Lemma 8.3, each window of six integers of s contributes two kept children; adversarial placement gives the mass bound $\sum_{j\geq 0}\big[2^{-c(6j+3.415)} + 2^{-c(6j+5.415)}\big]$, whose first two windows already sum to $1.0502 > 1$ at $c = 0.3$, with all such children within factor $2^{10.5}$ of the parent. Renewal induction with $A = 2^{-11c}$: for $X < 2^{11}y$, trivially $f(y, X) \geq 1 \geq A(X/y)^c$; for $X \geq 2^{11}y$, $f(y, X) \geq \sum_i f(y'_i, X) \geq AX^c \sum_i (y'_i)^{-c} \geq A(X/y)^c \cdot 1.0502$. Apply at the root (excluding the s=1 self-loop, absorbed by the worst-case tiling). $\square$

Remark 8.5 (position and refinement path)*.* The exponent sits between Crandall [2] and Krasikov's 1989 value 0.43; Krasikov–Lagarias reach 0.84175 by linear programs over difference-inequality systems mod 3^{11} [3]. The comparison is deliberate: the core above uses one door per state, two children per triple, and adversarial anchor phases; empirically the same core grows at exponent ≈ 0.45 and the full tree at ≈ 0.98. Each discarded resource corresponds to a stage of the Krasikov–Lagarias program (their residue systems are our door residues; their LP is an optimization over our branch inventory), with one structural difference: their inequalities *bound* the local branching, while Theorem 4.3 gives it *exactly*. Whether exactness improves on 0.84 is open and is claimed in neither direction.

9 Discussion

The mirror completes a symmetry the companion paper only gestured at: forward dynamics consume 2-adic digits under exact laws with no choice; backward dynamics offer infinite choice that moves only 3-adic data, with the 2-adic side frozen — yet the freeze itself routes complete control of the forward anchor through the choice of depth (Theorem 7.1(iii)). The conjecture, in these coordinates, is the assertion that a deterministic 2-adic flow and a controllable 3-adic tree are the same object. The asymmetries of Section 5 say the two sides are not interchangeable: the backward theory is in several respects *simpler* (fewer cross-prime steps, no class trichotomy, an unconditional valuation law), and the density program is the place where that simplicity is quantitatively testable against a mature literature. The concrete open items, in increasing ambition: reinstate doors, residue types, and anchor phases into the core of Theorem 8.4 and optimize (the LP stage); make the supercritical renewal analysis of Section 6 rigorous; and determine whether backward anchor controllability implies anything about the forward anchor walk.

A Methodology and records

This paper continues the protocol of [1, App. A]: the author directs and takes responsibility; the mathematical development is AI-assisted; every computational claim cites a runnable script; every result labeled proved was re-verified with independently written code before being so labeled. Two additions to the record here: one section (the four dual theorems of Section 5) was produced by a delegated session of a different model (Claude Sonnet) under a written brief with branch discipline, then independently re-verified in full (some 50,000 checks re-run) before merging; and two recorded corrections occurred during this paper's work — an initial renewal equation that ignored representative multiplicity, and a conjectured one-step window-steering property refuted by its own first test (the refutation became Theorem 7.1(ii)). The complete research record is public at `https://github.com/macindoe/collatz`.

References

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